



Volatility-Driven Asset Allocation: Integrating GARCH and ARIMA Models

A COMPARATIVE ANALYSIS: META-STRATEGY VS. MINIMUM VARIANCE PORTFOLIO
ON S&P 500

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Econometric Modeling & Volatility Estimation

Dual-Engine Framework:

GARCH(1,1) & ARIMA(1,0,0)

The core of the strategy lies in estimating conditional volatility by GARCH(1,1) model, which is essential for capturing volatility clustering effects.

Methodological Consistency & Weekly Frequency

To maintain transparency and rigor, all performance metrics are calculated on a weekly basis, matching the portfolio rebalancing frequency.

Robustness Mechanism:

The ARIMA Fallback To address GARCH convergence issues, often triggered by short estimation windows or low-volatility regimes

a structural fallback was implemented. If the GARCH estimation fails, the system automatically switches to the variance of ARIMA residuals. This hybrid approach ensures backtest continuity and provides stable risk estimates even during periods of extreme market stress

Part 1 – Risk Classification & Backtesting Framework

Forecasted vs. Historical Volatility

The primary objective is to determine if GARCH-based conditional volatility provides an informational advantage over a standard Historical volatility (standard deviation).

- GARCH (1,1): Captures dynamic market regimes and volatility clustering.
- Historical: Represents a static, backward-looking baseline for risk.

Robust Backtesting Design (Anti-Bias)

The analysis implements a rigorous walk-forward logic with weekly rebalancing to ensure real-world applicability:

- No Look-Ahead Bias: Only information available at the rebalancing date is used.
- No Survivorship Bias: The investable universe is dynamically adjusted using the historical composition of the S&P 500.
- Rolling Window: A 252-day window is used for continuous parameter estimation

Adaptive Risk Tiering (Tertile Classification)

Stocks are partitioned into three risk classes—Conservative, Moderate, and Aggressive—based on the tertiles of the volatility distribution:

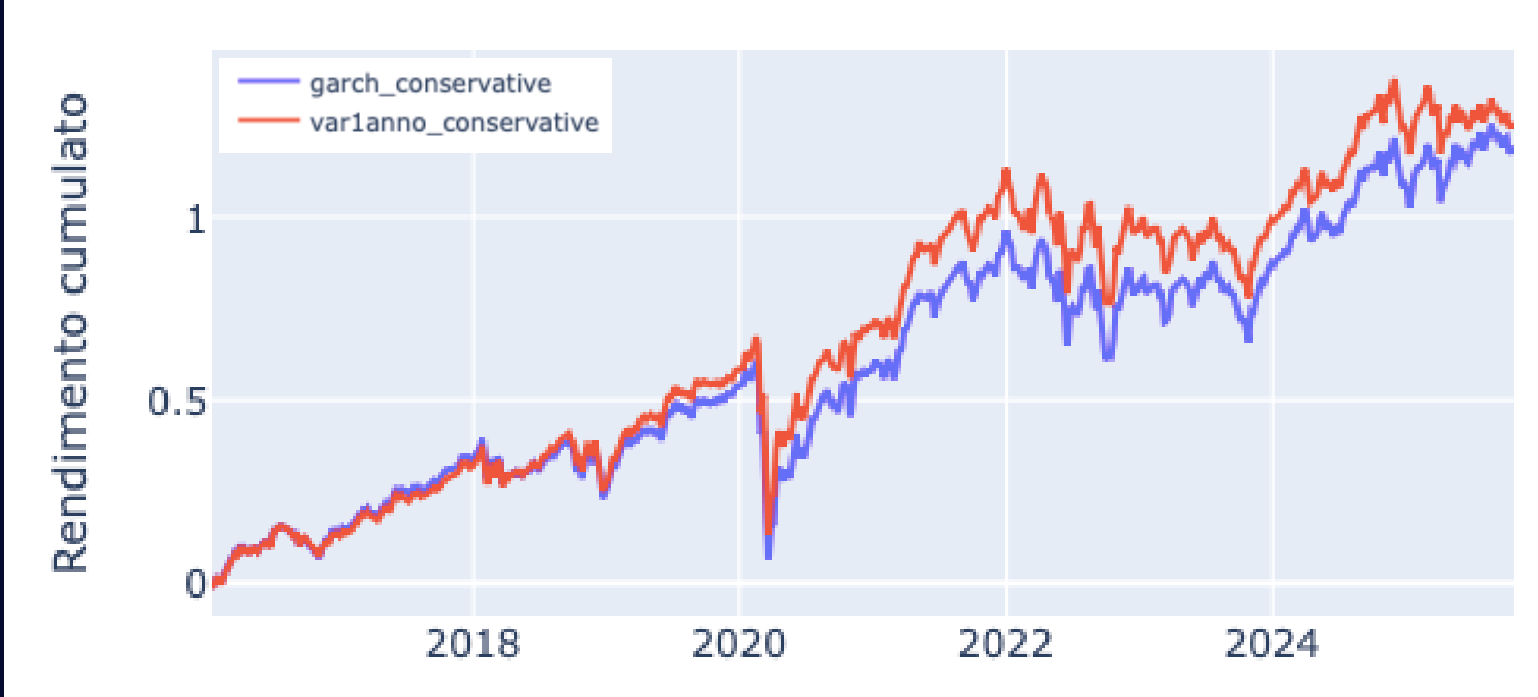
- Relative Classification: Tiers adapt to the current market context rather than using arbitrary fixed thresholds.
- Equal-Weight Portfolios: Six total strategies (3 GARCH-based, 3 Historical-based) are constructed using uniform weights to isolate the impact of the risk measure itself, avoiding optimizer-induced noise.

[252d Window] → [GARCH Estimation] → [Tertile Split] → [Weekly Rebalance]

Part 1 Analysis – The Conservative Portfolio

Metric (Weekly)	Historical Volatility	GARCH (1,1)
Avg. Weekly Return	0.00180	0.00175
Sharpe Ratio	0.0596	0.0574
Max Drawdown	-32.16%	-33.54%
Volatility	0.0206	0.0204

Performance Gap: Despite its econometric complexity, the GARCH(1,1) approach proves sub-optimal for low-risk assets, underperforming Historical Volatility in both Sharpe Ratio (0.0574 vs. 0.0596) and final cumulative return.



- Drawdown & Reactivity: The chart highlights that GARCH's high reactivity introduces "noise" during market shocks (e.g., the 2020 COVID crisis), leading to excessive rebalancing that amplifies the Maximum Drawdown (-33.54%) compared to the steadier historical measure (-32.16%).
- Conclusion: For capital preservation strategies, simple historical variance is more effective at capturing risk structures without the predictive instability inherent in conditional models within low-volatility market segments.

Part 1 Analysis – The Moderate Portfolio



- **Systematic Outperformance:** The GARCH strategy effectively captures market expansion phases, accumulating a persistent advantage over the historical counterpart, especially post-2020.
- **Drawdown Comparison:** Similar to the Conservative class, the GARCH portfolio experiences larger losses during extreme events (e.g., March 2020) due to its high reactivity to shocks.
- **Structural Differences:** The models select different assets, indicating that GARCH produces a genuine recomposition of the portfolio rather than a marginal adjustment.

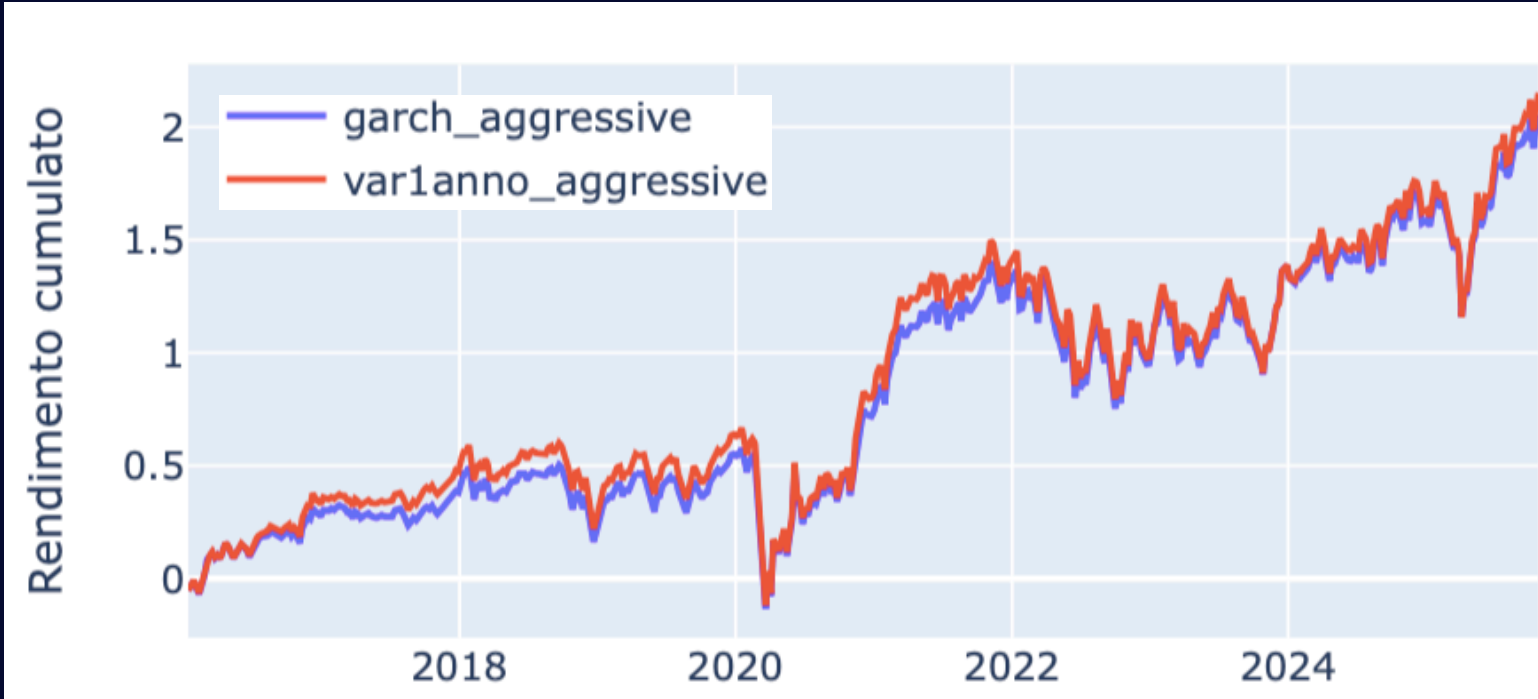
Metric (Weekly)	Historical Volatility	GARCH (1,1)
Avg. Weekly Return	0.00225	0.00240
Sharpe Ratio	0.0645	0.0691
Max Drawdown	-34.80%	-36.15%
Volatility	0.0260	0.0263

For moderate profiles, the ability of GARCH to ride recovery phases compensates for its higher drawdown, leading to more efficient wealth accumulation over time. Infact all the GARCH's performace are slightly higher

Part 1 Analysis – The Aggressive Portfolio

Metric (Weekly)	Historical Volatility	GARCH (1,1)
Avg. Weekly Return	0.00290	0.00281
Sharpe Ratio	0.064	0.063
Max Drawdown	47.2%	45.1%
Volatility	0.0362	0.0355

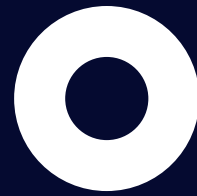
In the Aggressive tier, the Historical approach proves slightly more efficient over the long term. While GARCH offers a better Max Drawdown and quicker recovery, it fails to translate this reactivity into a persistent cumulative advantage.



- Path Irregularity: The GARCH portfolio exhibits wider fluctuations and greater sensitivity to shocks. The high reactivity of the model amplifies portfolio volatility without generating a structural return premium.
- Long-term Efficiency: The greater instability introduced by the GARCH model reduces the overall risk-return efficiency (Sharpe Ratio) compared to the steadier Historical measure
- For high-risk assets, the "noise" generated by constant re-estimation in GARCH offsets the benefits of its dynamic nature, making Historical Volatility a more robust for long-term growth.

Conclusions

Part 1



Key Findings

The analysis demonstrates that while GARCH(1,1) provides dynamic insights, its effectiveness is highly context-dependent. It does not universally improve performance across all risk profiles.

- Conservative Tier: Simpler historical measures are more effective for capital preservation.
- Moderate Tier: GARCH adds significant value by capturing growth during recovery phases.
- Aggressive Tier: The model introduces excessive "noise," making historical volatility a more robust .



Critical Issues of the GARCH Model

The study identifies two main structural weaknesses in an isolated GARCH-based strategy:

- Excessive Reactivity to Shocks: During extreme events, the model tends to overestimate future volatility, leading to inefficient "over-trading".
- Instability in Stress Phases: GARCH portfolios recorded deeper drawdowns compared to historical counterparts during systemic crises (e.g., March 2020)



Strategic improvement

To solve these limitations, the project evolves by introducing more robust allocation techniques:

- Class Integration: Combining Conservative, Moderate, and Aggressive tiers to balance the model's specific weaknesses.
- Dependence Measures: Moving beyond simple volatility to integrate Variance-Covariance matrices and Clustering to account for stock correlations and systematic risk.

Part 2 – From Volatility to Structural Selection

The Goal: Structural Diversification

The second phase of the project introduces a selection layer within the pre-established risk classes.

- Selection : After stocks are divided by GARCH volatility, the objective shifts to selecting an efficient subset of assets to maximize portfolio quality and diversification.
- Portfolio Efficiency: The focus moves from individual stock risk to the collective behavior of the assets, aiming to improve overall diversification.

The Var–Covar Selection Mechanism

The variance–covariance matrix of returns is utilized to explicitly model the dependence relationships between stocks within the same risk class.

- Diversification: Assets with identical volatility may exhibit high correlation (offering no protection) or negative correlation (providing portfolio benefits).
- Modeling Dependencies: While simple classification fails to distinguish these relationships, the var–covar matrix identifies which stocks contribute to actual risk reduction.

Implementation via Clustering

Clustering is applied to the correlation matrix, transformed into a distance matrix, to group stocks based on movement similarity.

- Structural Selection: Assets are selected from different clusters rather than based on individual rankings.
- Concentration Risk Mitigation: This process ensures that the portfolio is diversified across different market drivers, preventing over-exposure to specific systemic shocks within a GARCH risk class.

Part 2 Analysis – The Conservative Portfolio

Metric (Weekly)	GARCH Baseline (P1)	Var-Covar Selection	Best 30 (Return-based)
Avg. Weekly Return	0.0017	0.0019	0.0016
Weekly Volatility	0.0204	0.0207	0.0209
Sharpe Ratio	0.057	0.0616	0.054
Max Drawdown	-33.54%	-27.66%	-31.20%



The Var-Covar strategy achieves the highest risk-adjusted efficiency. The increase in returns is achieved with a volatility level nearly identical to the baseline, confirming that the outperformance is driven by selection quality.

Drawdown Containment: The reduction of the Max Drawdown from -33.54% to -27.66% is the most relevant empirical evidence in favor of using the correlation structure as a selection criterion.

Conclusion: For conservative profiles, the Var-Covar layer creates a more robust portfolio that is both more profitable and significantly more defensive during stress periods.

Part 2 Analysis – The Moderate Portfolio



In the Moderate class, the Var-Covar strategy significantly enhances efficiency. It achieves the highest Sharpe Ratio (0.0792) by improving returns while keeping volatility and drawdown aligned with the baseline, proving that even in this structural selection optimizes the risk-return trade-off

Efficiency without Added Risk: The increase in returns is not accompanied by higher volatility (0.0265 vs 0.0264). This indicates that the gain is derived from a systematic benefit of the dependence structure, not from higher exposure.

Conclusion: For moderate risk profiles, integrating GARCH classification with Var-Covar clustering provides a superior risk-adjusted growth path, maximizing efficiency without worsening the drawdown profile.

Metric (Weekly)	GARCH Baseline (P1)	Var-Covar Selection	Best 30 (Return-based)
Avg. Weekly Return	0.0024	0.0027	0.0023
Weekly Volatility	0.0264	0.0265	0.0263
Sharpe Ratio	0.0691	0.0792	0.0640
Max Drawdown	-36.40%	-36.01%	-36.62%

Part 2 Analysis – The Aggressive Portfolio

Metric (Weekly)	GARCH Baseline (P1)	Var-Covar Selection	Best 30 (Return-based)
Avg. Weekly Return	0.0028	0.0036	0.0034
Weekly Volatility	0.0355	0.0343	0.0338
Sharpe Ratio	0.0630	0.0875	0.0826
Max Drawdown	-45.10%	-33.73%	-31.80%



The Aggressive class showcases the most significant leap in efficiency. The Var-Covar selection achieves a vastly superior Sharpe Ratio (0.0875) while simultaneously lowering both volatility and Max Drawdown, proving that structural diversification is vital in high-risk segments

Market Outperformance: During expansion phases, the strategy exhibits faster growth, while corrections are noticeably less severe than the GARCH strategy. Infactthe Var-Covar selection reduces the Max Drawdown by nearly 10 percentage points compared to the baseline.

Conclusion: For aggressive profiles, the Var-Covar layer is a game-changer: it transforms high-volatility assets into a more regular and resilient return profile, significantly mitigating extreme losses.

Conclusions Part 2

1

The Added Value of Structural Selection

The integration of the Var-Covar matrix into the selection process provides a systematic improvement over the baseline GARCH portfolios.

- Performance Boost: Improvements in cumulative returns and Sharpe Ratios are consistent across all tiers, with the most pronounced effects in the Moderate and Aggressive classes.
- Risk Mitigation: In classes where stock interdependence is high, Var-Covar selection effectively reduces risk concentration and dampens extreme drawdowns.

2

Structural Analysis vs. Momentum (Best 30)

The comparison with the "Best 30" (selection based on past returns) highlights the necessity of a risk-centric approach.

- Lack of Robustness: While return-based selection may improve capital growth in specific phases, it proves less robust from a risk-return perspective.
- Consistency: The Var-Covar strategy provides a more stable and reliable advantage.

3

Final Synthesis:

The research confirms that the highest strategic value is achieved when two distinct dimensions are merged:

- Volatility Estimation (GARCH): Identifies the dynamic risk level of individual assets.
- Correlation-based Selection (Var-Covar): Optimizes the portfolio architecture through structural diversification.
- Outcome: This dual approach transforms a simple risk classification into a robust, and more efficient investment strategy.

Part 3 – Aggregated Portfolio

Objectives and Strategic Criteria

The third part of the research evaluates whether risk-based information translates into a systemic advantage at the aggregated portfolio level.

The methodology is guided by three core principles:

Simplicity: Reducing complexity to ensure the model's transparency.

Neutrality: Eliminating bias related to expected return forecasts.

Comparability: Maintaining a consistent framework to isolate the impact of risk modeling.

Methodology:

The adoption of Equal-Weight (EW) portfolios within each GARCH risk class is a deliberate choice:

Estimation Stability: It eliminates dependence on unstable and noisy estimates of expected returns.

Isolation of the GARCH Effect: By using equal weights, performance differentials are exclusively attributable to the stock selection criteria rather than allocation maneuvers.

Informational Clarity: This approach highlights the pure contribution of GARCH-based classification compared to traditional historical volatility.

Benchmark Selection:

Minimum Variance Optimization

To ensure a fair comparison, the market benchmark is constructed using a Minimum Variance optimization:

Risk-Based Consistency: Aligns with the overall project, as both the strategy and the benchmark rely solely on volatility information.

Efficiency: The Minimum Variance portfolio represents a natural solution on the Efficient Frontier for risk-averse objectives.

Neutral Comparison: Like the EW-GARCH approach, it avoids performance-based optimizations, removing the uncertainty associated with return forecasting.

Analysis Summary:

The aim of this framework is to demonstrate that any observed outperformance does not stem from aggressive allocation assumptions. Instead, it originates from a superior organization of risk information, proving the robustness of the project's underlying design.

Part 3 Analysis – Meta-Strategy vs. Benchmark

Metric (Weekly)	Minimum Variance (Benchmark)	Meta-Strategy
Avg. Weekly Return	0.0012	0.0027
Weekly Volatility	0.0159	0.0256
Sharpe Ratio	0.037	0.083
Max Drawdown	-21.81%	-33.08%



The set of metrics shows how the minimum variance portfolio effectively achieve the function of risk reduction, but at the cost of a significant compression of expected returns. Contrarily, the meta-strategy, while accepting greater return variability and more pronounced drawdowns, manages to achieve a markedly higher remuneration of risk, as demonstrated by the Sharpe ratio (meta-strategy Sharpe ratio ≈ 0.083 , minimum variance Sharpe ratio ≈ 0.038) and by average weakly returns.

The Meta-Strategy trajectory exhibits a distinct and widening decoupling from the benchmark. A significantly steeper slope is observed for the Meta-Strategy during market expansion phases, highlighting a superior ability to select assets with a higher growth profile."

General conclusions

1. Evidence from Analysis

The research confirms that an approach based on structured volatility and risk-class diversification is structurally more efficient than a purely defensive strategy.

Risk as Information: Volatility, when integrated with asset dependence (clustering), serves as a highly effective informational variable for robust portfolio construction.

Efficiency vs. Defense: The Minimum Variance portfolio successfully reduces risk but at the cost of significant return compression; the Meta-Strategy optimizes the risk-return trade-off.

2. Strategic Findings

Methodological Integrity: The outperformance is achieved without relying on expected return estimates, validating the project's focus on "Organized Risk Information".

Market Reactivity: The GARCH-based approach proves superior in exploiting expansion phases compared to traditional risk-minimization benchmarks.

3. Limitations & Future Extensions

Drawdown Sensitivity: Excessive reactivity of conditional volatility models can amplify losses during systemic crises in the absence of explicit exposure constraints.

Future Research: Potential extensions could involve integrating defensive constraints or implementing Fuzzy Logic for asset classification to mitigate extreme drawdowns.